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Clinical Efficacy of Ex Vivo Cross-Linking on Neovascularization of the Donor Cornea Carrier Tissue for the Boston Type I Keratoprosthesis

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Purpose: To evaluate the effect of corneal cross-linking (CXL) on corneal neovascularization (CNV) between eyes that were randomized to receive either CXL or non-CXL donor corneas as the carrier tissue for Boston keratoprosthesis surgery.

Design: A retrospective masked analysis of CNV from slit lamp photographs taken at postoperative weeks 16, 24, 36, and 52.

Subjects: Sixty-eight donor corneas were prospectively randomized 1:1 to receive either donor corneas that underwent ex vivo CXL or were non-CXL. The images of 47 corneas that were suitable for evaluation were included in the final analysis.

Methods: The slit lamp photos were analyzed morphometrically using a standardized protocol on Photoshop CS5 (Adobe Systems Inc) and ImageJ software (National Institutes of Health).

Main Outcome Measures: The 2 primary metrics used to quantify CNV were neovascular area (NA), defined as the area of corneal vessels projected onto the plane of a photograph, and invasion area (IA), defined as the fraction of corneal area in which vessels are present.

Results: Based on multivariable mixed-effects linear modeling, CXL reduces the percentage of NA in the CXL group by 2.2% ($P = 0.113$). Similarly, there is an average reduction of 7.8% in the percentage of IA in the CXL group compared with the non-CXL group ($P = 0.303$).

Conclusions: Although not statistically significant, this study observed a trend toward a lower CNV in CXL donor corneas compared with non-CXL donor corneas, suggesting that ex vivo CXL of donor corneas may protect against CNV of the donor cornea.

Financial Disclosure(s): Proprietary or commercial disclosure may be found in the Footnotes and Disclosures at the end of this article. *Ophthalmology Science 2025;5:100816 © 2025 Published by Elsevier Inc. on behalf of the American Academy of Ophthalmology. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).*

The normal cornea is devoid of blood and lymphatic vessels via antiangiogenic and antilymphangiogenic factors conferring its unique immune-privileged status.[1–3] However, this immunological privilege is disturbed by various inciting effects like hypoxia and inflammation, leading to corneal neovascularization (CNV). Corneal neovascularization can cause lipid deposition, tissue scarring, and chronic inflammation and is one of the main risk factors for immune-mediated corneal graft rejection after penetrating keratoplasty (PK).[2,4,5] The rejection process is believed to be started by migrating antigen-presenting cells to the regional lymph nodes via lymphatic vessels, causing accelerated graft sensitization.[1,3,4] This process is followed by effector cells that reach and reject the graft through the corneal blood vessels.[1,3,4] Therefore, regressing preexisting and preventing corneal blood and lymphatic vessels is crucial to improve the long-term survival rate in high-risk corneal transplants.

Collectively, preclinical data and case series have shown that corneal cross-linking (CXL) causes regression of CNV.[2] Furthermore, preclinical data have also demonstrated the role of CXL in preventing CNV.[2,4,6] A preclinical study investigated the effect of CXL pretreatment on lymphangiogenesis and hemangiogenesis. Their study found that CXL pretreatment significantly inhibits CNV by altering the structural stromal integrity and conferring increased extracellular matrix (ECM) stiffness that could halt the migration of vascular and lymphatic endothelial cells.[6]

While not originally intended as a therapy to prevent CNV, we recently completed a prospective, randomized, multicenter, double-masked, vehicle-controlled clinical trial

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https://doi.org/10.1016/j.xops.2025.100816
ISSN 2666-9145/25

that sought to determine whether ex vivo CXL of corneal tissue that served as a carrier for the Boston keratoprosthesis (BKPro) could prevent keratolysis (corneal melts) and improve retention of the prosthesis when compared with eyes that received tissue that did not undergo CXL.[7] The aforementioned preclinical finding that CXL can prevent CNV motivated us to perform masked analysis from our study's slit lamp photographs to determine whether eyes that received CXL tissue had less CNV than eyes that received non-CXL donor corneal tissue. Herein, we present the results of our masked image analysis of CNV between eyes that received CXL tissue or non-CXL tissue.

Methods

This study was a retrospective cohort designed to evaluate the effect of corneal CXL on CNV between eyes that were prospectively randomized to receive either CXL or non-CXL donor corneas as the carrier tissue for BKPro surgery. The clinical information and slit lamp photos were initially obtained as part of a phase I/II prospective, randomized, multicenter, double-masked, vehicle-controlled clinical trial, of which their methods and results have been previously published.[7] The study protocol was approved by the Mass General Brigham Institutional Review Board, Boston, Massachusetts. The study protocol adhered to the US Health Insurance Portability and Accountability Act and was performed in accordance with the tenets of the Declaration of Helsinki. This retrospective study utilized deidentified slit lamp photographs and did not involve any direct patient interaction or intervention. As such, individual informed consent was not required by the policies of our institutional review board and by the ethical standards for research involving minimal risk and deidentified data. The recruitment period spanned from April 21, 2017, to June 31, 2020. All surgeries were performed in the same time frame.

Patient Selection

Adult patients with corneal disease who were not candidates to receive donor keratoplasty but who were candidates for a BKPro with a high risk of keratolysis were included in the initial study. A high risk of keratolysis was defined as having a known underlying autoimmune disease or a history of previous sterile keratolysis in the operative eye. The subjects were randomized to receive either a CXL donor cornea tissue as a carrier for the BKPro or an untreated corneal tissue. The randomization scheme and surgical protocol have been published elsewhere.[7] To ensure balanced stratification between the 2 arms of the study, the randomization scheme was stratified by center and preoperative diagnostic inclusion criteria, which included the presence of an autoimmune disease or prior sterile keratolysis.[7] Briefly, cross-linking was performed 2 days before the surgical date using a modified Dresden protocol.[8–10] Briefly, corneas were de-epithelialized, and a topical riboflavin solution (0.1% riboflavin and 20% dextran) was applied at intervals of 2 to 3 minutes over a span of 15 minutes. Subsequently, ultraviolet light was applied for 30 minutes, with additional drops of riboflavin applied every 5 minutes throughout the procedure. Riboflavin exposure causes the tissue to develop a slightly yellow tint, which persists even after thorough rinsing. To ensure consistency, non-CXL corneas underwent de-epithelialization and received the same riboflavin treatment but were not exposed to ultraviolet light.[7] Storage of corneas followed standard care procedures, and the donor corneas were placed in a sterile container containing storage solution (Optisol-GS; Bausch & Lomb) before and after the CXL. In addition, the endothelium was bathed in the same type of storage media throughout the CXL using the anterior chamber maintainer.[7] Masked surgeons adhered to established protocols for the assembly and implantation of the BKPro type 1 with an 8.5-mm-diameter backplate.[11] Postoperative management was established before enrollment and remained consistent across all patients. Further details on postoperative management have been published elsewhere.[7]

Slit Lamp Photo Analysis

A retrospective masked analysis of CNV from slit lamp photographs from participants' eyes that were randomized 1:1 to receive carrier tissue for BKPro surgery that either underwent ex vivo CXL or non-CXL was performed. Slit lamp photographs taken at postoperative weeks 16, 24, 36, and 52 were analyzed. Images that met the following quality standards were included in the final analysis: no glare, in focus, and in a linear alignment with the objective lens.

The slit lamp photos were analyzed morphometrically using a standardized protocol used in published clinical trials and open-label studies.[5,12,13] The slit lamp photos were analyzed morphometrically using Photoshop CS5 (Adobe Systems Inc) as the graphic editing software and ImageJ software (National Institutes of Health). The central optic was the same size for each eye studied, and the area of the cornea analyzed was standardized according to the backplate diameter. First, the limbus was marked and cropped with an elliptical marquee tool to denote the total area of the cornea. Then, the blood vessels were enhanced and highly saturated for easier identification, and by using a freeform pen tool, they were traced from the limbus to the center of the cornea. With Photoshop filters, selection tools, and layers, a threshold level was set, and the nonvessel area was erased. Finally, the distal ends of the longest vessels were connected if they were in 2-clock-hour proximity (Fig 1). Using ImageJ software, the images were analyzed in a binary mode.

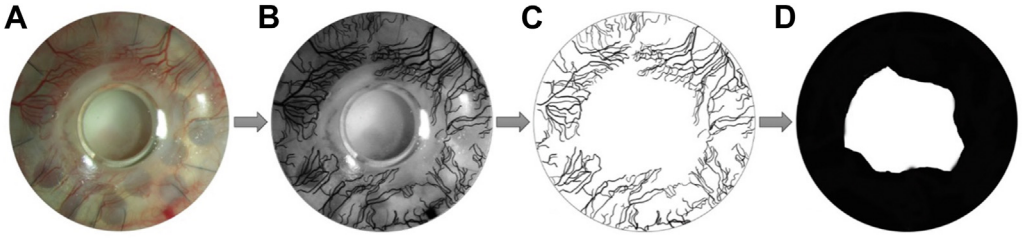


Figure 1. Morphometrical analysis of slit lamp photos. **A,** The limbus was marked to denote the total area of the cornea. **B,** The corneal blood vessels were traced starting from the limbus. **C,** The nonvessel area was erased by setting a threshold level (neovascular area). **D,** The distal ends of the vessels were connected (invasion area).

Quantification of CNV

The 2 primary metrics used to quantify CNV were neovascular area (NA) and invasion area (IA), as previously described.[5,12] Neovascular area is defined as the area of the corneal vessels themselves when projected into the plane of a photograph.[5,12,13] For this, the blood vessels were traced, and by setting a threshold level, the nonvessel area was erased. The remaining NA was then pixelized and measured. The calculated NA was normalized to the whole corneal area to obtain the NA percentage.[13] The second metric was IA, defined as the fraction of corneal area in which vessels are present.[5,12,13] The vessels were traced from the limbus to the center of the cornea, and the very ends of all vascular sprouts were marked and connected if they were in 2-clock-hour proximity, and the IA was traced. The measured area was again normalized to the whole corneal area to obtain the IA percentage.[13]

Statistical Analysis

A Mann−Whitney *U* test and a mixed-effects linear model with repeated measures were used to assess the differences between the 2 metrics. A multivariable analysis was performed to compare CXL versus control, adjusting for age, sex, race, and autoimmune disease. A 2-tailed $P < 0.05$ was considered statistically significant. Analysis was performed using Stata software version 18 (StataCorp LLC).

Results

Sixty-eight subjects were prospectively randomized 1:1 in the study between 2017 and 2020. Forty-seven corneas had slit-lamp photographs that met our quality standards for morphometric analysis: 26 corneas in the CXL group and 21 in the non-CXL group. Demographic and baseline clinical information of these 2 groups is presented in Table 1. The mean age of patients in the CXL group was 64.9 years (interquartile range = 28−87), whereas, in the non-CXL group, it was 64.9 years (interquartile range = 29−89). Both age and sex did not significantly differ between groups.

Masked observers analyzed slit lamp photographs of the cornea taken at 4 different time points over 52 weeks. They measured the NA to quantify the area of the cornea covered by the vessels and the IA to quantify the fraction of the cornea that contains vessels. In general, neovascularization within the donor cornea increased over time in both study groups, as demonstrated by large increases in the percentage of NA and IA. For instance, the percentage of NA increased from a median of 3.2% at week 16 to 7.7% at week 52 for the non-CXL group and from 3.3% to 4.5% during the same period for the CXL treatment group (Table 2). Similarly, during the same interval, the percentage of IA increased from a median of 34.7% to 48.9% for the non-CXL group and from 37.6% to 38% for the CXL group (Table 2).

When comparing the treatment arms, the median NA and the IA were less for the CXL group than the non-CXL group at nearly all time points; however, these differences were not statistically significant at the individual time points as measured by the Mann−Whitney *U* test (Table 2). The percentage of NA at weeks 24, 36, and 52 were 3.4%, 2.2%, and 4.5%, respectively, in the CXL group in comparison to values of the non-CXL group (8.1%, 4.8%, and 7.7%, respectively) (Table 2). Regarding the percentage of IA, it was >30% less in the CXL versus the non-CXL group at week 24 and between 7% and 10% less in weeks 36 and 52 (Table 2).

To determine whether there was a significant treatment effect over time, a multivariable mixed-effects linear modeling was used, adjusted for age, sex, race, and autoimmune disease; the fixed effects being the treatment group (CXL or non-CXL), random effects pertaining to patient, and the repeated measures being the percentage of NA and IA at the 4 time points. Adjusting for age, sex, race, and

Table 1. Baseline Characteristics

Demographic Characteristics	Total	Group		P Value
		Non-CXL (n = 21)	*CXL (n = 26)*	
Age, mean (range)*	64.9 (28−89)	64.9 (29−89)	64.9 (28−87)	0.997
Sex, n (%)†				
Male	34 (72.3)	16 (76.2)	18 (69.2)	0.596
Female	13 (27.7)	5 (23.8)	8 (30.8)	
Race, n (%)†				
White	30 (63.8)	14 (66.7)	16 (61.5)	0.436
Black	7 (14.9)	4 (19.1)	3 (11.5)	
Asian	2 (4.3)	0 (0)	2 (7.7)	
Native American	2 (4.3)	0 (0)	2 (7.7)	
Other	6 (12.8)	3 (14.3)	3 (11.5)	
Hispanic/Latino ethnicity, n (%)†	5 (10.7)	2 (9.5)	3 (11.5)	0.999
Preoperative autoimmune diagnosis, n (%)	10	5 (50)	5 (50)	
Stevens−Johnson syndrome		2	2	
Rheumatoid arthritis		2	1	
Graft-versus-host disease		1	1	
Ocular cicatricial pemphigoid		0	1	

CXL = corneal cross-linking.
*T test.
†Chi-square test.

Table 2. Percentage of Vessel Area and Percentage of Invasion Area between Intervention Groups at Different Time Points

Time Point and Outcome	Intervention		P Value
	Non-CXL (*n* = 21)	*CXL* (*n* = 26)	
Week 16			
Percentage of neovascular area	3.2 (1.4–9.9)	3.3 (1.8–7)	0.839
Percentage of invasion area	34.7 (11.6–55.9)	37.6 (17.3–71.4)	0.598
Week 24			
Percentage of neovascular area	8.1 (3.8–14.3)	3.4 (1–8.9)	0.093
Percentage of invasion area	73.6 (38.4–93.9)	34.7 (8.2–74.1)	0.085
Week 36			
Percentage of neovascular area	4.8 (2–8)	2.2 (1.1–6.3)	0.104
Percentage of invasion area	51.8 (36.2–62.5)	43.1 (15.1–53)	0.104
Week 52			
Percentage of neovascular area	7.7 (3–9.9)	4.5 (2.5–6.4)	0.248
Percentage of invasion area	48.9 (37.8–57.7)	38 (19.7–57.1)	0.364

CXL = corneal cross-linking.
Quantitative values are presented with median and interquartile range, and *P* values are calculated using the Mann–Whitney *U* test.

autoimmune disease, the mixed-effects linear model for a percentage of NA, CNV reduction of 2.2% over time, was observed ($P = 0.113$) (Table 3). Similarly, a mixed-effects linear model for the dependent variable percentage of IA found a decrease in the IA of 7.8% over time ($P = 0.303$) (Table 3).

Discussion

Corneal neovascularization is one of the main risk factors for immune-mediated corneal graft rejection after PK.[1] Penetrating keratoplasties performed as part of BKPro implantation may be at even higher risk of CNV, given the advanced underlying pathologies and the fact that the addition of the prosthetic device may incite neovascularization. Hence, several therapeutic strategies to induce regression of CNV have been studied, including VEGF inhibitors and CXL, with reported variable effectiveness.[4] Given that the study of therapeutic strategies to prevent CNV is more limited than those aimed at treating CNV and that emerging evidence from preclinical and clinical studies has demonstrated that CXL can prevent CNV,[4,6] we were motivated to analyze images from our recently published study, as identifying therapeutic alternatives to prevent CNV in BKPro patients could significantly improve long-term graft survival and visual outcomes.[7] Therefore, this retrospective cohort study evaluated the effect of corneal CXL on CNV between eyes that were prospectively randomized to receive either CXL or non-CXL donor corneas as carriers for the BKPro. Our study showed that, in general, neovascularization increases over time in both intervention groups. Our study suggests that CXL pretreatment may be an effective preventive strategy to reduce CNV as indicated by a decrease of 7.8% over time in the percentage of IA ($P = 0.36$).

Our study's results support the findings of preclinical and clinical studies. For example, a preclinical randomized study in rats evaluated the effect of CXL pretreatment followed by suture placement on angiogenesis.[6] The later study found that CXL pretreatment significantly suppresses CNV by decreasing lymphangiogenesis and hemangiogenesis.[6] Similar results were also reported in a murine model that found that preoperative CXL in vascularized high-risk

Table 3. Comparison of Percentage of Neovascular Area and Percentage of Invasion Area According to Group, Based on Multivariable Mixed-Effects Linear Modeling with Repeated Measures

Outcome and Group	Adjusted Coefficient	Marginal Mean (SE)	95% Confidence Interval		P Value
			Lower	*Upper*	
Percentage of neovascular area*					
Non-CXL (reference)	−2.2	6.8 (1.1)	−5.1	0.6	0.113
CXL		4.6 (0.9)			
Percentage of invasion area†					
Non-CXL (reference)	−7.8	48.1 (5.5)	−22.7	7.1	0.303
CXL		39.7 (4.9)			

CXL = corneal cross-linking; SE = standard error.
Multivariable models are adjusted for age, sex, race, and autoimmune disease as covariates.
*Dependent variable: percentage of neovascular area.
†Dependent variable: percentage of invasion area.

eyes can significantly improve subsequent graft survival.[2] Additionally, a retrospective clinical study evaluated the effects of peripheral CXL before PK or combined with PK or central CXL before PK in vascularized high-risk eyes. This study also showed that peripheral CXL significantly reduced CNV without significant revascularization and no immune reactions.[4]

Prior published studies found that CXL effectively treats CNV by suppressing vascular endothelial elongation, immune cell proliferation, and infiltration.[2] Moreover, CXL can cause regression of CNV by the reaction of riboflavin and ultraviolet A light that releases reactive oxygen radicals, which causes apoptosis of keratocytes and vascular endothelial cells in the corneal stroma, targeting corneal blood and lymphatic vessel extension.[2,14,15] Regarding preventive therapy, pretreatment with CXL is known to alter the structural stromal integrity and increase the ECM density.[6] The ECM density determines both corneal biomechanical stiffness and fibrovascular behavior.[6,16] The increased biomechanical stiffness obstructs the inflammatory cell infiltration and halts stem cell migration, proliferation, and differentiation, resulting in less lymphangiogenesis and angiogenesis.[6,17] Under inflammatory conditions, ECM is proteolytically altered, which may decrease resistance to angiogenesis. However, CXL pretreatment could counteract the proteolytic alteration that occurs under inflammatory conditions by increasing stiffness to the matrix and inhibiting the migration of vascular and lymphatic endothelial cells.[18,19]

This study has several limitations, such as those associated with any retrospective studies.[20,21] Despite identifying a clinically important trend, the small sample size of the study limited our ability to reach statistical significance in either of the study parameters between the 2 groups at any specific time interval after surgery or using a mixed-effects linear model with repeated measure. The original study was not powered to study the effect that CXL has on CNV.[7] Moreover, due to the retrospective design of the study, slit lamp photographs at earlier time points are not available. As a result, the analysis does not capture the early effects of CXL on CNV, which prior studies have shown to be more pronounced in the initial weeks after treatment.[22] Furthermore, given the retrospective design of the study, some patients were lost to follow-up, leading to a retention rate of 42.6% at the 52-week follow-up.

Despite the assembly and implantation of the BKPro following established standards and the postoperative eye drop management regimen being established before the beginning of the trial, other factors such as suture removal were not consistent for all eyes. The lack of a block randomization for many of the potential confounding variables could be considered another study limitation. For instance, the underlying pathology of the nonautoimmune eyes was not known; however, the eyes were randomized using permuted block randomization using a 1:1 allocation ratio according to (1) autoimmune disease or prior sterile keratolysis and (2) study site. Therefore, it is important to note that the randomization scheme used decreased the probability that a particular uncontrolled variable (e.g., underlying diagnosis for the nonautoimmune eyes, surgeon timing of suture removal, site photography, or size of the graft) confounded the study results for one of the treatment groups.

Moreover, since the slit lamp photographs were initially obtained as part of a prospective randomized study, their quality standards could not be modified.[7] Therefore, images were taken with different slit lamps, different cameras, different lighting settings, and different angulations. Moreover, our study used a subjective step to measure neovascularization, leading to a step that is difficult to reproduce and time-consuming. Furthermore, the images were taken from above the eye using "en face" slit lamp photographs, which, along with the limitations of working with 2-dimensional pictures, can make it difficult to accurately assess the plane of the vessels or stromal edema.[23] Additionally, external factors such as light reflex can obscure corneal vessel tracing; therefore, we cannot rule out the possibility of detecting retrocorneal neovascularization in some cases.

In summary, the trend of a reduction in the percentage of NA and percentage of IA in response to CXL pretreatment suggests that CXL of donor corneas may confer resistance to CNV and provides additional evidence to other preclinical and clinical studies that found CXL prevents CNV and therefore could be an adjunctive measure to improve graft survival.[2,4,6] These results suggest that ex vivo CXL of donor tissue may be a particularly promising option to improve the survival rates of keratoplasties that are defined as high risk due to pathological corneal blood vessels that are present in the recipient bed and are known to lower graft survival rates.[1] Future research, including large-scale prospective randomized clinical studies, is needed to establish the safety and efficacy of ex vivo CXL of donor corneas in eyes at high risk for rejection due to CNV, as well as to assess the potential additional use of CXL in situ after BKPro to not only stabilize the cornea but also regress CNV.

Footnotes and Disclosures

Originally received: September 24, 2024.
Final revision: April 14, 2025.
Accepted: April 24, 2025.
Available online: May 1, 2025. Manuscript no. XOPS-D-24-00395.

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Disclosures:

All authors have completed and submitted the ICMJE disclosures form.

The authors made the following disclosures:

J.B.C.: Financial support – Generously supported by the Blavatnik Family Foundation.

This study was supported by the United States Department of Defense's Defense Medical Research and Development, grant number W81XWH-15-2-0044. In-kind support (riboflavin solution and UV light source [KXL system]) was received from Avedro Inc, a wholly owned subsidiary of Glaukos Corporation. The funding organization had no role in the design or conduct of this research.

HUMAN SUBJECTS: No human subjects were included in this study. The study protocol was approved by the Mass General Brigham Institutional Review Board, Boston, Massachusetts. The study protocol adhered to the US Health Insurance Portability and Accountability Act and was performed in accordance with the tenets of the Declaration of Helsinki. This retrospective study utilized deidentified slit lamp photographs and did not involve any direct patient interaction or intervention. As such, individual informed consent was not required by the policies of our institutional review board and by the ethical standards for research involving minimal risk and deidentified data.

No animal subjects were used in this study.

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Obtained funding: N/A

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Abbreviations and Acronyms:
BKPro = Boston keratoprosthesis; **CNV** = corneal neovascularization; **CXL** = corneal cross-linking; **ECM** = extracellular matrix; **IA** = invasion area; **NA** = neovascular area; **PK** = penetrating keratoplasty.

Keywords:
Cross-linking, Corneal neovascularization, Keratoprosthesis.

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